Athulith Paraselli

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Portfolio: https://aparaselli.github.io/

Research Interests

Mechanistic Interpretability, Causal Abstraction, Machine Cognition

Education

Brown University, Providence, RI — Sep 2025 - May 2027
Master of Science, Computer Science - Artificial Intelligence/Machine Learning — GPA: 4.00 / 4.00

University of California, San Diego (UCSD), La Jolla, CA — Sep 2021- June 2025
Bachelor of Science, Mathematics-Computer Science, *Data Science Minor* — GPA: 3.82 / 4.00

Research Experience

Serre Lab, Brown University — Jun 2026 – Present
Advised by Dr. Thomas Serre
- Discovered a sparse set of attention heads that mediate conjunctive/disjunctive visual search within VLMs.
- Applied Representational Similarity Analysis (RSA) to analyze internal visual representations across transformer layers in VLMs.

LUNAR Lab, Brown University — Sep 2025 – Present
Advised by Dr. Ellie Pavlick
- Identified and characterized a "modality gap" within context-memory conflict resolution, discovering how VLMs inconsistently recall facts when presented with conflicting information.
- Engineered a generalized mechanistic interpretability pipeline for VLMs, implementing attribution patching and logit lens to trace information flow across modalities.
- Established causal links between model components and behavior using back-patching and attention masking, successfully inducing and mitigating the modality gap phenomenon.

Machine Learning, Perception, and Cognition Lab, UCSD — Apr 2025 – Aug 2025
Advised by Dr. Xi Li and Dr. Zhuowen Tu
- Developed novel multimodal adversarial training pipeline to improve robustness of VLMs by 12%.
- Integrated LoRA and prompt tuning in a unified training loop, enabling flexible defense strategies.

Sugihara Lab, UCSD — Nov 2024 – Nov 2025
Advised by Dr. George Sugihara
- Developed an LSTM-based forecasting method for partially observed dynamical systems using state-space reconstruction to improve prediction of chaotic ecological data.
- Fine-tuned Chronos, a foundation model for time-series forecasting, on ecological data to reduce weighted quantile loss (wQL) of algal bloom predictions by 32%.

Technical Experience

Software Engineer Intern, Qualcomm — Jun 2024 – Sep 2024
Modem Integration Testing
- Created an efficient unit testing workflow through optimizing existing industry-standard methods.
- Implemented functionality to automatically generate unit tests based on customer specifications.

Publications

Slow to See, Slow to Suppress: Understanding the Effects of Modality in Context-Memory Conflicts
Athulith Paraselli, Tianze Hua, and Ellie Pavlick
Findings of EMNLP 2026

Working Papers

Predicting Bioluminescent Algal Blooms by Conditioning Neural Forecasts on Reconstructed Dynamics
Athulith Paraselli, Ciro Zhang, Melissa L. Carter, and George Sugihara
Under review at the NeurIPS 2026 Workshop on AI for Stochastic Dynamics

Presentations and Posters

Understanding the Effects of Modality in Context-Memory Conflicts
Athulith Paraselli, Tianze Hua, Ellie Pavlick
The 3rd New England Mechanistic Interpretability Workshop, 2026

Predicting Bioluminescent Algal Blooms Using Hybrid "EDM-LSTMs"
Athulith Paraselli, Ciro Zhang, Esther Chung, Nian-Nian Wang
San Diego Undergraduate Tech Conference, 2025

Teaching Experience

Undergraduate Instructional Assistant
Conduct exam review sessions, design homework/exams, hold weekly office hours, grade assignments
Principles of Data Science, *UCSD* — Apr 2024 – Jun 2025
Advanced Machine Learning Methods, *UCSD* — Apr 2025 – Jun 2025

Services & Outreach

Eta Kappa Nu (HKN) Projects Program — Nov 2024 – Jun 2025
Inaugural Project Lead
- Led a team of 4 students to develop a live data processing pipeline and website to display Bioluminescent algal bloom forecasts.
- Established connections between Labs at Scripps Institution of Oceanography and Eta Kappa Nu.
- Coordinated community outreach initiatives, presenting accessible explanations of Harmful Algal Blooms and sharing updates from ongoing research collaborations.
- Presented results at San Diego Undergrad Tech Conference.

Eta Kappa Nu (HKN) Events Team — Mar 2024 – Jun 2025
Math-CS Events Chair
- Organized and led technical workshops on fundamental ML concepts such as neural networks and gradient descent, enhancing accessibility for underclassmen.
- Collaborated with board members to coordinate events, recruit inductees, and promote technical engagement across the department.

Technical Skills

Programming Languages
Python, C/C++, Java, Perl, ARM Assembly, MATLAB, SQL, HTML, CSS, JavaScript, Svelte

Areas of Expertise
Version Control(Git/Github), Machine Learning(Scikit-learn, Pytorch, Tensorflow, Keras, Hugging Face, NNSight), AWS(Amplify, API Gateway, Lambda, DynamoDB)

Awards & Honors

- Finalist, CBAI Summer Research Fellowship in AI Safety — 2026
- Eta Kappa Nu (IEEE HKN) Member, *UCSD* — 2023-2025
- Provost Honors x11, *UCSD* — 2021-2025